

Activity__Explore confidence intervals

May 12, 2026

1 Activity: Explore confidence intervals

1.1 Introduction

The Air Quality Index (AQI) is the Environmental Protection Agency's index for reporting air quality. A value close to 0 signals little to no public health concern, while higher values are associated with increased risk to public health. The United States is considering a new federal policy that would create a subsidy for renewable energy in states observing an average AQI of 10 or above.

You've just started your new role as a data analyst in the Strategy division of Ripple Renewable Energy (RRE). **RRE operates in the following U.S. states: California, Florida, Michigan, Ohio, Pennsylvania, Texas.** You've been tasked with constructing an analysis which identifies which of these states are most likely to be affected, should the new federal policy be enacted.

Your manager has requested that you do the following for your analysis: 1. Provide a summary of the mean AQI for the states in which RRE operates. 2. Construct a boxplot visualization for AQI of these states using `seaborn`. 3. Evaluate which state(s) may be most affected by this policy, based on the data and your boxplot visualization. 4. Construct a confidence interval for the RRE state with the highest mean AQI.

1.2 Step 1: Imports

1.2.1 Import packages

Import `pandas` and `numpy`.

```
[1]: # Import relevant packages

### YOUR CODE HERE ###

import pandas as pd
import numpy as np
```

1.2.2 Load the dataset

The dataset provided gives national Air Quality Index (AQI) measurements by state over time. Pandas is used to import the file `c4_epa_air_quality.csv` as a DataFrame named `aqi`. As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

Note: For the purposes of your analysis, you can assume this data is randomly sampled from a larger population.

```
[2]: # RUN THIS CELL TO IMPORT YOUR DATA

### YOUR CODE HERE ###
aqi = pd.read_csv('c4_epa_air_quality.csv')
```

1.3 Step 2: Data exploration

1.3.1 Explore your dataset

Before proceeding to your deliverables, spend some time exploring the `aqi` DataFrame.

```
[4]: # Explore your DataFrame `aqi`.

### YOUR CODE HERE ###

aqi.describe(include='all')
print(aqi.describe())

print(aqi['state_name'].value_counts())
```

```
<bound method NDFrame.describe of      Unnamed: 0  date_local
state_name      county_name \
0          0  2018-01-01      Arizona      Maricopa
1          1  2018-01-01         Ohio      Belmont
2          2  2018-01-01      Wyoming      Teton
3          3  2018-01-01  Pennsylvania  Philadelphia
4          4  2018-01-01         Iowa      Polk
..          ...      ...
255         255  2018-01-01  District Of Columbia  District of Columbia
256         256  2018-01-01      Wisconsin      Dodge
257         257  2018-01-01      Kentucky      Jefferson
258         258  2018-01-01      Nebraska      Douglas
259         259  2018-01-01   North Carolina      Wake

      city_name      local_site_name \
0      Buckeye      BUCKEYE
```

1	Shadyside	Shadyside
2	Not in a city	Yellowstone National Park - Old Faithful Snow ...
3	Philadelphia	North East Waste (NEW)
4	Des Moines	CARPENTER
..	...	...
255	Washington	Near Road
256	Kekoskee	HORICON WILDLIFE AREA
257	Louisville	CANNONS LANE
258	Omaha	NaN
259	Not in a city	Triple Oak

	parameter_name	units_of_measure	arithmetic_mean	aqi
0	Carbon monoxide	Parts per million	0.473684	7
1	Carbon monoxide	Parts per million	0.263158	5
2	Carbon monoxide	Parts per million	0.111111	2
3	Carbon monoxide	Parts per million	0.300000	3
4	Carbon monoxide	Parts per million	0.215789	3
..	...	...	...	...
255	Carbon monoxide	Parts per million	0.244444	3
256	Carbon monoxide	Parts per million	0.200000	2
257	Carbon monoxide	Parts per million	0.163158	2
258	Carbon monoxide	Parts per million	0.421053	9
259	Carbon monoxide	Parts per million	0.188889	2

[260 rows x 10 columns]>

California	66
Arizona	14
Ohio	12
Florida	12
Texas	10
New York	10
Pennsylvania	10
Michigan	9
Colorado	9
Minnesota	7
New Jersey	6
Indiana	5
North Carolina	4
Massachusetts	4
Maryland	4
Oklahoma	4
Virginia	4
Nevada	4
Connecticut	4
Kentucky	3
Missouri	3
Wyoming	3
Iowa	3

Hawaii	3
Utah	3
Vermont	3
Illinois	3
New Hampshire	2
District Of Columbia	2
New Mexico	2
Montana	2
Oregon	2
Alaska	2
Georgia	2
Washington	2
Idaho	2
Nebraska	2
Rhode Island	2
Tennessee	2
Maine	2
South Carolina	1
Puerto Rico	1
Arkansas	1
Kansas	1
Mississippi	1
Alabama	1
Louisiana	1
Delaware	1
South Dakota	1
West Virginia	1
North Dakota	1
Wisconsin	1

Name: state_name, dtype: int64

Question: What time range does this data cover?

This data covers all of the data collected on January 1st, 2018.

Question: What are the minimum and maximum AQI values observed in the dataset?

The minimum and maximum AQI values observed in the dataset are 0 and 50 respectively.

Question: Are all states equally represented in the dataset?

No, all states are not equally represented in the dataset. Some states have more sites than others.

Hint 1

Refer to [the content about descriptive statistics](#).

Hint 2

Use `pandas` or `numpy` to explore the `aqi` DataFrame.

Hint 3

Use any of the following functions: - pandas: `describe()`,`value_counts()`,`shape()` - numpy: `unique()`,`mean()`

1.4 Step 3: Statistical tests

1.4.1 Summarize the mean AQI for RRE states

Start with your first deliverable. Summarize the mean AQI for the states in which RRE operates (California, Florida, Michigan, Ohio, Pennsylvania, and Texas).

```
[5]: # Summarize the mean AQI for RRE states.

### YOUR CODE HERE ###

# Create a list of RRE states.

rre_states = ['California','Florida','Michigan','Ohio','Pennsylvania','Texas']

# Subset `aqi` to only consider these states.

aqi_rre = aqi[aqi['state_name'].isin(rre_states)]

# Find the mean aqi for each of the RRE states.

aqi_rre.groupby(['state_name']).agg({"aqi":"mean","state_name":"count"}) #alias_
→as aqi_rre
```

```
[5]:
```

	aqi	state_name
state_name		
California	12.121212	66
Florida	5.500000	12
Michigan	8.111111	9
Ohio	3.333333	12
Pennsylvania	2.900000	10
Texas	2.700000	10

Hint 1

Subset your DataFrame to only include those states in which RRE operates.

Hint 2

Define a list consisting of the states in which RRE operates and use that list to subset your DataFrame.

Hint 3

Use pandas `isin()` to subset your DataFrame by the list of RRE states.

1.4.2 Construct a boxplot visualization for the AQI of these states

Seaborn is a simple visualization library, commonly imported as `sns`. Import `seaborn`. Then utilize a boxplot visualization from this library to compare the distributions of AQI scores by state.

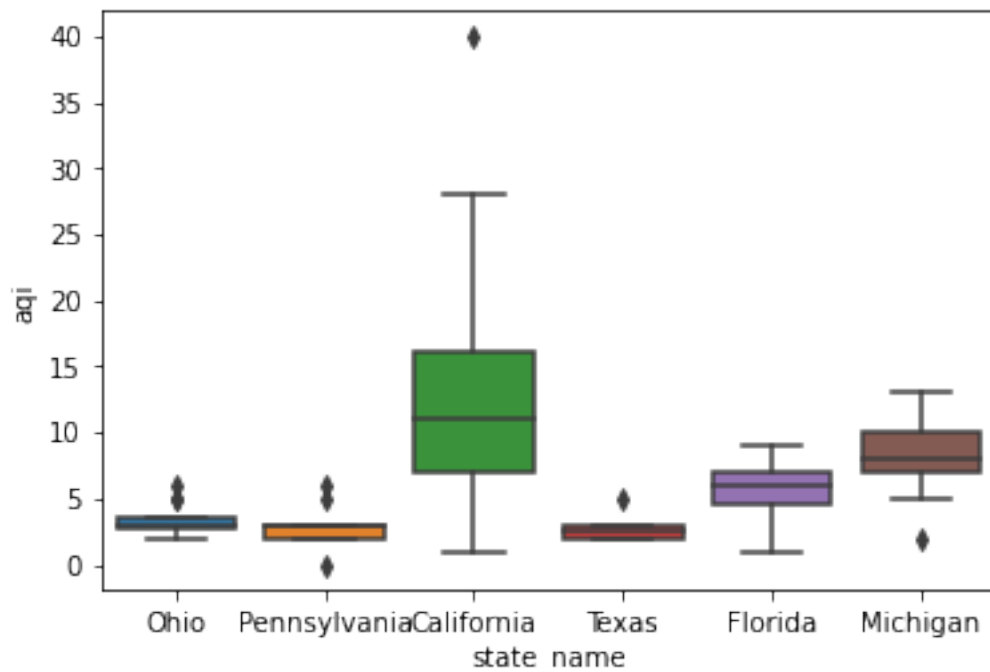
```
[6]: # Import seaborn as sns.  
  
### YOUR CODE HERE ###  
  
import seaborn as sns
```

1.4.3 Create an in-line visualization showing the distribution of aqi by state_name

Now, create an in-line visualization showing the distribution of aqi by `state_name`.

```
[7]: ### YOUR CODE HERE ###  
  
sns.boxplot(x=aqi_rre["state_name"],y=aqi_rre["aqi"])
```

```
[7]: <matplotlib.axes._subplots.AxesSubplot at 0x7e17af57cd50>
```



Hint 1

Use the boxplot visual for this purpose.

Hint 2

Reference [Seaborn's boxplot visualization documentation](#).

Hint 3

Assign `state_name` to the x argument and `aqi` to the y.

Question: Based on the data and your visualizations, which state(s) do you suspect will be most affected by this policy?

I suspect California and Michigan will be the most affected states by this policy based on the boxplots' means and ranges.

Hint 1

Consider the mean AQI for the RRE states, as well as the distribution in the boxplots relative to the policy limit (10).

1.4.4 Construct a confidence interval for the RRE state with the highest mean AQI

Recall the 4-step process in constructing a confidence interval:

1. Identify a sample statistic.
2. Choose a confidence level.
3. Find the margin of error.
4. Calculate the interval.

1.4.5 Construct your sample statistic

To construct your sample statistic, find the mean AQI for your state.

```
[12]: # Find the mean aqi for your state.

### YOUR CODE HERE ###

aqi_mo = aqi[aqi['state_name']=='Missouri']

sample_mean = aqi_mo['aqi'].mean()
sample_mean
```

```
[12]: 3.0
```

Hint 1

Reference what you've previously learned to recall what a [sample statistic](#) is.

Hint 2

Calculate the mean for your highest AQI state to arrive at your sample statistic.

Hint 3

Call the `mean()` function within `pandas` on your DataFrame.

1.4.6 Choose your confidence level

Choose your confidence level for your analysis. The most typical confidence level chosen is 95%; however, you can choose 90% or 99% if you want decrease or increase (respectively) your level of confidence about your result.

```
[10]: # Input your confidence level here:
```

```
### YOUR CODE HERE ###
```

```
confidence_level = 0.95
```

```
confidence_level
```

```
[10]: 0.95
```

1.4.7 Find your margin of error (ME)

Recall **margin of error** = $z \times \text{standard error}$, where z is the appropriate z -value for the given confidence level. To calculate your margin of error:

- Find your z -value.
- Find the approximate z for common confidence levels.
- Calculate your **standard error** estimate.

Confidence Level	Z Score
90%	1.65
95%	1.96
99%	2.58

```
[13]: # Calculate your margin of error.
```

```
### YOUR CODE HERE ###
```

```
# Begin by identifying the z associated with your chosen confidence level.
```

```
z_value = 1.96
```

```
# Next, calculate your standard error.
```

```
standard_error = aqi_mo['aqi'].std() / np.sqrt(aqi_mo.shape[0])
```

```
print("standard error:")
```

```
print(standard_error)
```

```
# Lastly, use the preceding result to calculate your margin of error.
```

```
margin_of_error = standard_error * z_value
```



```
print("margin of error:")
print(margin_of_error)
```

```
standard error:
1.0
margin of error:
1.96
```

1.4.8 Calculate your interval

Calculate both a lower and upper limit surrounding your sample mean to create your interval.

```
[14]: # Calculate your confidence interval (upper and lower limits).

### YOUR CODE HERE ###

upper_ci_limit = sample_mean + margin_of_error
lower_ci_limit = sample_mean - margin_of_error
(lower_ci_limit, upper_ci_limit)
```

```
[14]: (1.04, 4.96)
```

Hint 1

Refer to [the content about constructing a confidence interval](#).

Hint 2

Identify the sample mean from your prior work. Then use the margin of error to construct your upper and lower limits.

Hint 3

Subtract the margin of error from the sample mean to construct your lower limit, and add the margin of error to your sample mean to construct your upper limit.

1.4.9 Alternative: Construct the interval using `scipy.stats.norm.interval()`

`scipy` presents a simpler solution to developing a confidence interval. To use this, first import the `stats` module from `scipy`.

```
[15]: # Import stats from scipy.

### YOUR CODE HERE ###

from scipy import stats
```

1.5 Step 4: Results and evaluation

1.5.1 Recalculate your confidence interval

Provide your chosen `confidence_level`, `sample_mean`, and `standard_error` to `stats.norm.interval()` and recalculate your confidence interval.

```
[16]: ### YOUR CODE HERE ###

stats.norm.interval(alpha=confidence_level, loc=sample_mean,
    ↪ scale=standard_error)
```

```
[16]: (1.040036015459946, 4.959963984540054)
```

2 Considerations

What are some key takeaways that you learned from this lab?

Based on the mean AQI observations, California and Michigan were most likely to have a mean AQI above 10. California had the highest sample mean AQI which means that it could be the state most likely to be affected by the policy change.

Other than these, I learned more about confidence intervals and how they could allow me to estimate the sample mean with a level of confidence.

What findings would you share with others?

I would share the process of me getting to the conclusions, especially elaborating on the confidence interval, along with the inference from the data.

What would you convey to external stakeholders?

I would share with them that California had the highest observed AQI, and would suggest them to focus on that state. I would also share the result of the 95% confidence interval, and convey any potential shortcomings of this analysis.

References

[seaborn.boxplot](#) — [seaborn 0.12.1 documentation](#). (n.d.).

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.